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Brillouin Scattering and Relaxation in Liquids

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The spectrum of light inelastically scattered from density fluctuations is considered both for nonrelaxing and relaxing fluids. When the thermal conductivity of the liquid is small, the functions describing these spectra take on particularly simple forms. In addition to the unshifted Rayleigh line, caused by scattering from isobaric density fluctuations, the spectrum for a nonrelaxing liquid is just the response of a resonant system driven by thermal energy. For a relaxing liquid, the spectral distribution function has a somewhat more complicated form, but its interpretation is still quite straightforward. The use of electric circuit equivalents illustrates the principal features of the response spectrum. These results are formal ones, but while they are useful in understanding the scattering process, they are not particularly suited for the analysis of experimental results. For this purpose, approximations are developed that enable one to relate the outstanding characteristics of the spectrum (for example, the positions and half-widths of shifted lines) to fundamental relaxation parameters, such as the longitudinal storage and loss moduli.

INTRODUCTION

THE use of laser light sources and high-resolution detection techniques has made it feasible to experimentally probe the frequency spectrum of light scattered by thermally driven fluctuations in fluid media.¹⁻⁷ The shape of this spectrum is determined by such quantities as the transport coefficients, elastic moduli and relaxation times of the fluid. Therefore, an analysis of experimentally determined spectra can yield valuable information concerning these parameters of the medium.

In recent years, a large body of data relating to the relaxational behavior of liquids has been accumulated predominantly through ultrasonic propagation studies. The highest frequency available in such an experiment is on the order of 10^9 Hz so that this technique is useful

only for studying phenomena on time scales longer than 10^{-10} sec. This has been a serious limitation in the investigation of relaxation processes in liquids of the Kneser type in which the relaxation involves the coupling of energy between internal and external degrees of freedom of the fluid. In addition, the study of structural relaxation has been limited to a few special classes of liquids. Among these are such complicated liquids as highly bonded hydrogen-bonded liquids,⁸⁻¹⁰ some inorganic glasses and molten salts,¹¹⁻¹³ and some very long chain polymer liquids.¹⁴⁻¹⁶ Because of the need for relatively high viscosities, relaxational data could be obtained only in the supercooled region where the molecular motions are highly cooperative. The use of reduced variables and reduced functions to estimate the high-temperature noncooperative behavior of these

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¹ G. B. Benedek, J. B. Lastovka, K. Fritsch, and T. Greytak, *J. Opt. Soc. Am.* **54**, 1284-1285 (L) (1964).

² R. Y. Chiao and B. P. Stoicheff, *J. Opt. Soc. Am.* **54**, 1286-1287 (L) (1964).

³ T. J. Greytak and G. B. Benedek, *Proc. IEEE* **53**, 1293-1299 (1965); *Phys. Rev. Letters* **17**, 179-185 (1966).

⁴ P. A. Fleury and R. Y. Chiao, *J. Acoust. Soc. Am.* **39**, 751-752 (L) (1966).

⁵ D. I. Mash, V. S. Starunov, and I. L. Fabelinskii, *Soviet Phys.—JETP* **20**, 523-524 (1965).

⁶ D. H. Rank, E. M. Kiess, U. Fink, and T. A. Wiggins, *J. Opt. Soc. Am.* **55**, 925-927 (1965); **56**, 163-166 (1966).

⁷ W. S. Gornall, G. I. A. Stegeman, B. P. Stoicheff, R. H. Stolen, and V. Volterra, *Phys. Rev. Letters* **17**, 297-299 (1966).

⁸ K. F. Herzfeld and T. A. Litovitz, *Absorption and Dispersion of Ultrasonic Waves* (Academic Press Inc., New York, 1959), Chap. 12.

⁹ T. A. Litovitz and C. M. Davis, in *Physical Acoustics*, W. P. Mason, Ed. (Academic Press Inc., New York, 1965), Vol. 2, Part A, Chap. 5.

¹⁰ R. Meister, C. J. Marhoffer, R. Sciamanda, L. Cotter, and T. A. Litovitz, *J. Appl. Phys.* **31**, 854-870 (1960).

¹¹ P. B. Macedo and T. A. Litovitz, *Phys. Chem. Glasses* **6**, 69-80 (1965).

¹² J. D. Tauke, P. B. Macedo, and T. A. Litovitz (to be published).

¹³ G. J. Gruber and T. A. Litovitz, *J. Chem. Phys.* **40**, 13-26 (1964).

¹⁴ J. D. Ferry, *Viscoelastic Properties of Polymers* (John Wiley & Sons, Inc., New York, 1961).

¹⁵ W. Philipoff, Ref. 9, Part B, Chap. 7.

¹⁶ R. S. Marvin and J. E. McKinney, Ref. 9, Part B, Chap. 9.

liquids provides no real solution, for, without adequate high-frequency data, such estimates are little more than extrapolations of the low-temperature behavior.

It is desirable to obtain viscoelastic data in simpler fluids. The difficulty is that, because of their high fluidity, these liquids possess such short structural relaxation times that ultrasonic techniques are inadequate to probe the nature of the structural relaxation process in them. However, the use of light-scattering experiments makes possible the study of molecular processes for time intervals as short as 10^{-11} sec; this is equivalent to extending the available frequency range from 10^9 Hz to greater than 10^{10} Hz. This increased frequency range is a critical one for the study of thermal relaxation since estimates indicate that many liquids have a vibrational relaxation time of the order of 10^{-11} sec.

In order to obtain the desired viscoelastic data from the scattered light spectrum, some model of the behavior of the fluctuations which cause the scattering is needed. Many derivations of the scattered light spectrum exist in the literature.¹⁷⁻²⁶ Recently, Mountain^{27,28} has employed hydrodynamic models^{29,30} to obtain theoretical spectra for two classes of fluids. In one case, the generalized structure factor $S(\kappa, \omega)$, which gives the spectral distribution,³¹⁻³³ was obtained for a simple monoatomic fluid with frequency-independent transport coefficients. In the second case, this result was generalized to describe fluids in which a relaxing specific heat leads to a frequency-dependent bulk viscosity. This latter treatment is applicable to any single relaxation process, no matter what its origin.

In many liquids, the assumption of single relaxation is untenable; the results obtained by Mountain,²⁸ however, indicate what type of information one can obtain even in these cases. For example, the spectral distribution depends upon the thermal conductivity and specific heats of the medium and upon such viscoelastic properties as the "zero" and "infinite" frequency moduli or sound velocities and the distribution of relaxation times. Detailing the actual procedure by which these parameters can be obtained from measurements of the spectrum of scattered light is the chief aim of this paper.

I. SPECTRUM OF THE SCATTERED LIGHT

When linearly polarized light is scattered by a fluid, the scattered light, consisting of both a polarized and a depolarized part,^{24-26,34} shows a distribution in frequency about the value appropriate for the incident beam. Moreover, the light scattered through an angle θ measured with respect to the undeflected beam is shifted in wave vector by an amount κ , the magnitude of which is given by

$$\kappa = |\mathbf{k}' - \mathbf{k}_i| = 2nk_i \sin(\theta/2), \quad (1)$$

where $\mathbf{k}'$ and $\mathbf{k}_i$ are, respectively, the wave vectors of the scattered and incident light and n is the refractive index of the medium. If ω represents the shift in angular frequency of the scattered light, then the spectrum of the scattered light observed at angle θ can be described by a function $S(\kappa, \omega)$.

The spontaneous local variations in the dielectric constant which are responsible for the scattering can be considered as being caused by fluctuations in local temperature, density and molecular orientations. Because of the weak explicit dependence of the molecular polarizability on temperature, the effect of the first of these can be neglected. And since the fluid is isotropic, it can also be assumed that orientational and density fluctuations are uncorrelated. The spectral function $S(\kappa, \omega)$ is thus composed of two parts: $S_{II}(\kappa, \omega)$, which describes the spectrum of scattered light with polarization parallel to that of the incident beam, and $S_I(\kappa, \omega)$, which describes the spectrum of light with polarization normal to it. The depolarized portion of the spectrum $S_I(\kappa, \omega)$ is caused solely by anisotropy in the molecular polarizability tensor; orientational fluctuations produce variations in this tensor causing the depolarized scattering. The polarized spectrum $S_{II}(\kappa, \omega)$ contains a contribution from these orientational fluctuations in addition to an isotropic part caused by density fluctuations. The isotropic contribution to the scattering is given by the generalized structure factor $S(\kappa, \omega)$ which is related to the polarized and depolarized spectra by

$$S(\kappa, \omega) = S_{II}(\kappa, \omega) - fS_I(\kappa, \omega), \quad (2)$$

where f is a geometrical factor the value of which is determined by symmetries in the molecular polarizability tensor. If, as is often the case for organic molecules, there is one axis of symmetry, i.e., two equal eigenvalues of the tensor, the value of f can be shown

³⁴ D. A. Pinnow, S. J. Candau, and T. A. Litovitz (to be published).

- ¹⁷ L. Brillouin, *Ann. Phys. (Paris)* **17**, 88-122 (1922).
- ¹⁸ M. Leontovich, *J. Phys. (U.S.S.R.)* **4**, 499-514 (1941).
- ¹⁹ J. Frenkel, *Kinetic Theory of Liquids* (Oxford University Press, London, 1946), Chap. 5.
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- ²¹ C. I. Carr and B. H. Zimm, *J. Chem. Phys.* **18**, 1616-1626 (1950).
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- ²⁵ R. Pecora and W. A. Steele, *J. Chem. Phys.* **42**, 1872-1879 (1965).
- ²⁶ V. S. Starunov, *Opt. Spectry*, **19**, 176-170 (1966).
- ²⁷ R. D. Mountain, *Rev. Mod. Phys.* **38**, 205-214 (1966).
- ²⁸ R. D. Mountain, *J. Res. Natl. Bur. Std.* **70A**, 207-220 (1966).
- ²⁹ L. D. Landau and E. M. Lifshitz, *Fluid Mechanics* (Addison-Wesley Publ. Co., Reading, Mass., 1959), Chap. 17.
- ³⁰ S. M. Rytov, *Soviet Phys.—JETP* **6**, 130-140, 401-408, 513-522 (1958).
- ³¹ L. Van Hove, *Phys. Rev.* **95**, 249-262 (1954).
- ³² L. I. Komarov and I. Z. Fisher, *Soviet Phys.—JETP* **16**, 1358-1361 (1963).
- ³³ H. A. Stuart, *Die Struktur des Freien Moleküls* (Springer Verlag, Berlin, 1952).

to be $4/3$.³³ For a full discussion of depolarized light scattering, the reader is referred to the works of Pecora^{24,25} and of Pinnow, Candau, and Litovitz.³⁴ In this paper, only the isotropic spectrum $S(\kappa, \omega)$ is discussed; for this problem, the vector nature of κ can be suppressed.

Following Mountain,²⁸ it is convenient to define a spectral distribution function $\sigma(\kappa, \omega)$ by

$$\sigma(\kappa, \omega) = S(\kappa, \omega) / S(\kappa) \quad (3)$$

where $S(\kappa)$ is the ordinary structure factor

$$S(\kappa) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega S(\kappa, \omega). \quad (4)$$

For light scattering in liquids away from the critical point, the scale of distances is large compared with the range of correlation, and the frequencies involved are small compared with the frequencies of molecular vibrations, so that one may use a continuum approximation and employ macroscopic variables in treating the problem. The appropriateness of such an approach has been discussed by Sugawara and Yip.³⁵ In this continuous medium limit $S(\kappa, \omega)$ is just the double Fourier transform of the autocorrelated density.³¹ Thus

$$\sigma(\kappa, \omega) = \langle \rho(\kappa, \omega) \rho(-\kappa) \rangle / \langle \rho(\kappa) \rho(-\kappa) \rangle, \quad (5)$$

where the brackets $\langle \dots \rangle$ indicate an ensemble average over initial states of the system. In this expression,

$$\langle \rho(\kappa, \omega) \rho(-\kappa) \rangle = \left(\frac{2}{\pi} \right)^{1/2} \int_0^{\infty} dt e^{-i\omega t} \langle \rho(\kappa, t) \rho(-\kappa) \rangle, \quad (6)$$

where

$$\rho(\kappa, t) = (2\pi)^{-1/2} \int d\mathbf{r} e^{i\kappa \cdot \mathbf{r}} \rho(\mathbf{r}, t) \quad (7)$$

and

$$\rho(\kappa) = \rho(\kappa, t) |_{t=0}. \quad (8)$$

The quantity $\rho(\mathbf{r}, t)$ in Eq. 7 is the departure from its equilibrium value of the density at time t . The desired correlation function $\langle \rho(\kappa, t) \rho(-\kappa) \rangle$ can be constructed in the following way. Consider a large collection of thermodynamically identical systems. The density in these systems will fluctuate randomly about its thermodynamic value. We examine each of the systems beginning at the moment when the Fourier component of the density fluctuation has some specific value $\rho(\kappa)$. A time t later the fluctuations for the various systems are $\rho_1(\kappa, t)$, $\rho_2(\kappa, t)$, etc. Define the conditional average $[\rho(\kappa, t)]_{\text{av}}$ as the average of these $\rho_i(\kappa, t)$ taken subject to the condition

$$\rho_i(\kappa, 0) = \rho(\kappa), \quad i = 1, 2, \dots \quad (9)$$

This average is then assumed to obey a set of "macroscopic equations" descriptive of the particular model being considered. After these equations are solved for $[\rho(\kappa, t)]_{\text{av}}$ the function $\langle [\rho(\kappa, t)]_{\text{av}} \rho(-\kappa) \rangle$ can be con-

structed. But since $\langle \dots \rangle$ indicate an average over all initial states of the system, the above conditional averaging need not be explicitly considered: that is, $\langle [\rho(\kappa, t)]_{\text{av}} \rho(-\kappa) \rangle = \langle \rho(\kappa, t) \rho(-\kappa) \rangle$. In what follows, then, we write simply $\rho(\kappa, t)$, understanding by this the average defined above.

The macroscopic equations will give $\rho(\kappa, t)$ in terms of $\langle \rho(\kappa) \rho(-\kappa) \rangle$ and some additional quantities characterizing the history of the system. Such quantities are the initial values of time derivatives of ρ . These can be specified by suitable thermodynamic averages.

For those cases to be treated in this paper, $\langle \rho(\kappa, t) \rho(-\kappa) \rangle$ will be found to depend upon $\langle \rho(\kappa) \rho(-\kappa) \rangle$ and $\langle \dot{\rho}(\kappa) \rho(-\kappa) \rangle$, where $\dot{\rho}(\kappa)$ is the Fourier space transform of the initial value of the first time derivative of the density fluctuation. The first of these can be found using equilibrium statistical mechanics:

$$\langle \rho(\kappa) \rho(-\kappa) \rangle = \rho_0^2 k_B T \beta_T / v_\kappa, \quad (10)$$

where ρ_0 is the equilibrium density, T the equilibrium temperature, k_B is Boltzmann's constant, β_T is the isothermal compressibility, and v_κ is the volume of the fluctuation. For the κ th Fourier component this volume is of the order κ^{-3} . Because specifying a value of $\rho(\kappa)$ in no way specifies a value of $\dot{\rho}(\kappa)$, we assume, after Mountain,

$$\langle \dot{\rho}(\kappa) \rho(-\kappa) \rangle = 0, \quad (11)$$

that is, $\dot{\rho}$ and ρ are considered statistically independent. This statement is just one of thermodynamic equilibrium in the fluid; it means just that there is no mass transport taking place in the fluid. This follows for we can show by integration by parts that

$$\langle \dot{\rho}(\kappa) \rho(-\kappa) \rangle \propto \lim_{t_1 \rightarrow \infty} [\rho(\kappa, t) \rho(-\kappa, t)]_{t=t_1}^{t=t_1},$$

which is zero for a stationary process.

To construct $\sigma(\kappa, \omega)$, we shall follow the procedure suggested by Landau and Placzek³⁶ and employed by Mountain.^{27,28} For the time scale of interest, assume that the density fluctuations are governed by the linearized hydrodynamic equations. For a nonrelaxing fluid, these are the continuity equation

$$\partial \rho / \partial t + \rho_0 \nabla \cdot \mathbf{u} = 0, \quad (12)$$

the longitudinal part of the Navier-Stokes equation

$$\frac{\partial \mathbf{u}}{\partial t} = -\frac{c_0^2}{\rho_0 \gamma} \nabla \rho - \frac{c_0^2 \varphi}{\gamma} \nabla T + \left(\frac{\eta_0}{\rho_0} \right) \nabla (\nabla \cdot \mathbf{u}), \quad (13)$$

and the energy transport equation

$$\frac{\partial T}{\partial t} = \left(\frac{\gamma - 1}{\rho_0 \varphi} \right) \frac{\partial \rho}{\partial t} + \left(\frac{\lambda}{\rho_0 C_v} \right) \nabla^2 T. \quad (14)$$

In these equations, $\mathbf{u}$ is the particle velocity, ρ_0 is the equilibrium fluid density, ρ is the fluctuation in density, and T is the temperature fluctuation. The low-fre-

³⁶ L. Landau and G. Placzek, *Physik Z. Sowjetunion* 5, 172-173 (1934).

³⁵ A. Sugawara and S. Yip, *Phys. Fluids* 10, 1911-1921 (1967).

quency sound velocity is c_0 , the thermal expansion coefficient is φ , and λ is the thermal conductivity. The ratio of the specific heat at constant pressure C_p to the specific heat at constant volume C_v is γ , and η_0 is the longitudinal viscosity

$$\eta_0 = \frac{4}{3}\eta_s + \eta_v, \quad (15)$$

η_s and η_v being the shear and volume viscosities, respectively. Formally, η_v can be used to represent any loss mechanism other than shear viscosity. Mountain²⁷ obtains the solution to these equations by Laplace transforming on time and Fourier transforming on spatial variables. This specifies that the problem is to be solved in an infinite medium subject to the given

initial conditions. The spectral distribution function can then be obtained by observing that

$$\sigma(\kappa, \omega) = 2 \operatorname{Re}\{[\langle \rho(\kappa, s) \rho(-\kappa) \rangle / \langle \rho(\kappa) \rho(-\kappa) \rangle]_{s=i\omega}\}, \quad (16)$$

where $\rho(\kappa, s)$ is the Laplace (time) transform and Fourier (space) transform of $\rho(\mathbf{r}, t)$. After carrying out the indicated transforms, solving for $\rho(\kappa, s)$, and multiplying through by $\rho(-\kappa)$, we take the ensemble average of the result. Terms linear in $\langle \rho(\kappa) \rho(-\kappa) \rangle$, $\langle \dot{\rho}(\kappa) \rho(-\kappa) \rangle$, and $\langle T(\kappa) \rho(-\kappa) \rangle$ appear. Using Eq. 11 the second of these may be put to zero. Similarly, the term in $\langle T(\kappa) \rho(-\kappa) \rangle$ vanishes because of the statistical independence of density and temperature fluctuations. The term remaining yields

$$\frac{\langle \rho(\kappa, s) \rho(-\kappa) \rangle}{\langle \rho(\kappa) \rho(-\kappa) \rangle} = \frac{s^2 + [(\lambda/\rho_0 C_v) + (\eta_0/\rho_0)] s \kappa^2 + c_0^2 \kappa^2 (1 - 1/\gamma) + \lambda \eta_0 \kappa^4 / \rho_0^2 C_v}{s^3 + [(\lambda/\rho_0 C_v) + (\eta_0/\rho_0)] s^2 + [c_0^2 \kappa^2 + (\lambda \eta_0 \kappa^4 / \rho_0^2 C_v)] s + \lambda c_0^2 \kappa^4 / \gamma \rho_0 C_v}. \quad (17)$$

Denote the denominator of Eq. 17 by $D(s)$. For transparent fluids, the thermal conductivity is usually quite small; that is

$$\lambda/C_v \ll \eta_0. \quad (18)$$

Thus we can neglect terms of order $(\lambda/C_v \eta_0)^2$ and factor $D(s)$ approximately as

$$D(s) \approx \left(s + \frac{\lambda \kappa^2}{\rho_0 C_p} \right) \left\{ s^2 + \left[\frac{\eta_0}{\rho_0} + \frac{\lambda(1-1/\gamma)}{\rho_0 C_v} \right] \kappa^2 s + c_0^2 \kappa^2 \right\}. \quad (19)$$

For most liquids, even the term $\lambda(1-1/\gamma)/\rho_0 C_v$ may be neglected in comparison with η_0/ρ_0 . This has been discussed by Biquard²⁷ in connection with the absorption of ultrasound where $\eta_0/\rho_0 + \lambda(1-1/\gamma)/\rho_0 C_v$ is proportional to the absorption coefficient. Except for liquid metals, which are not amenable to light scattering studies anyway, $\lambda(1-1/\gamma)/\rho_0 C_v$ amounts to only a few tenths of a percent of η_0/ρ_0 . It is therefore neglected in Eq. 19. Defining

$$\Gamma = \eta_0 \kappa^2 / 2\rho_0 \quad (20)$$

and

$$\omega_0 = c_0 \kappa, \quad (21)$$

Eq. 19 becomes

$$D(s) = (s + \lambda \kappa^2 / \rho_0 C_p) (s^2 + 2\Gamma s + \omega_0^2). \quad (22)$$

To this point, the treatment is identical with that of Mountain²⁷; in what follows, a decomposition and interpretation of the spectrum which is substantially different from his is presented. This procedure is instructive as it leads to further insight into the scattering process.

Using Eq. 22 in Eq. 17 and decomposing the right

side into partial fractions gives

$$\frac{\langle \rho(\kappa, s) \rho(-\kappa) \rangle}{\langle \rho(\kappa) \rho(-\kappa) \rangle} = \left(1 - \frac{1}{\gamma} \right) \frac{1}{s + (\lambda \kappa^2 / \rho_0 C_p)} + \left(\frac{1}{\gamma} \right) \frac{s + 2\Gamma}{(s + \Gamma)^2 + (\omega_0^2 - \Gamma^2)}. \quad (23)$$

The spectral distribution of the scattered light is then

$$\sigma(\kappa, \omega) = \left(1 - \frac{1}{\gamma} \right) \frac{2(\lambda \kappa^2 / \rho_0 C_p)}{(\lambda \kappa^2 / \rho_0 C_p)^2 + \omega^2} + \left(\frac{2c_0^2 \kappa^2}{\gamma} \right) \frac{2\Gamma}{(2\omega\Gamma)^2 + (\omega^2 - \omega_0^2)^2}. \quad (24)$$

The first term in this expression can be identified as the normal central or Rayleigh line caused by entropy fluctuations at constant pressure. The second term represents the Mandelshtam-Brillouin doublet caused by adiabatic pressure fluctuations. It is useful to denote this term by $\sigma_s(\kappa, \omega)$ and to consider it separately. Combining it with Eqs. 5 and 10 leads to

$$\sigma_s(\kappa, \omega) \propto \langle \rho_s^2(\kappa, \omega) \rangle = \frac{2(\rho_0^2 / v_s) k_B T \eta_0}{(\omega \eta_0)^2 + (\rho_0 \omega^2 / \kappa^2 - M_0)^2}, \quad (25)$$

where

$$M_0 = \rho_0 c_0^2 \quad (26)$$

is the adiabatic bulk modulus. Observe that the form of $\langle \rho_s^2(\kappa, \omega) \rangle$ is the same as that for the spectrum of the displacement correlation function for a harmonic system under the action of a random (white noise) driving force.^{38,39}

³⁸ G. E. Uhlenbeck and L. S. Ornstein, *Phys. Rev.* **36**, 823-841 (1930).

³⁹ L. D. Landau and E. M. Lifshitz, *Statistical Physics* (Addison-Wesley Publ. Co., Reading, Mass., 1958), Chap. 12.

²⁷ P. Biquard, *Ann. Phys. (Paris)* **6**, 195-304 (1936).

This equivalence of the spectra of spontaneous fluctuations with that of a randomly driven system is at first surprising. The surprise comes about because the terms *spontaneous* and *driven* are often taken as exclusive; the fact is, of course, that spontaneous fluctuations occur because they are *driven* by random forces associated with the thermal bath to which the system is coupled. If one could remove this coupling—i.e., remove the thermal forces—one would, in so doing, eliminate the spontaneous fluctuations.

However, it seems, that mathematically no account of these forces has been taken—Eqs. 12–14 governing the dynamics of the situation are all homogeneous. This difficulty is resolved when one realizes that the effect of the forces *has* been included through the imposition of initial (equilibrium) conditions. The particular choice of initial conditions made here, Eqs. 10 and 11, is equivalent to assuming that the density fluctuations are driven by a force whose power spectrum is white noise.

The spectral density function $\sigma_s(\kappa, \omega)$ exhibits peaks at

$$\omega_p = \pm \omega_0 [1 - \frac{1}{2}(\omega_0 \eta_0 / M_0)^2]^{\frac{1}{2}}. \quad (27)$$

The situation is completely analogous to the problem of the damped harmonic oscillator of natural frequency ω_0 and damping coefficient η_0 . The frequency of free oscillation is $\omega_0 [1 - \frac{1}{4}(\omega_0 \eta_0 / M_0)^2]^{\frac{1}{2}}$ while the resonant frequency of the driven system is $\omega_0 [1 - \frac{1}{2}(\omega_0 \eta_0 / M_0)^2]^{\frac{1}{2}}$. It is, thus, the “resonant” frequency of the density fluctuations driven by the thermal energy of the system, rather than the oscillation frequency of freely oscillating temporally decaying fluctuations, that is represented by the peaks in the Brillouin spectrum.

This might have been anticipated, for in the limit of small thermal conductivity, the linearized hydrodynamic equations can be combined to give the differential equation

$$\left(\frac{\rho_0}{\kappa^2}\right) \frac{\partial^2}{\partial t^2} \rho_s(\kappa, t) + \eta_0 \frac{\partial}{\partial t} \rho_s(\kappa, t) + M_0 \rho_s(\kappa, t) = 0 \quad (28)$$

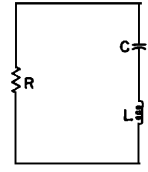
for the κ th Fourier component of the adiabatic density fluctuations. The desired spectral distribution $\langle \rho_s^2(\kappa, \omega) \rangle$ then follows as a straightforward application of the fluctuation-dissipation theorem of Callen.^{40,41} This theorem relates the fluctuations of a physical quantity in an equilibrium system to the dissipative process associated with the response of that system to an external driving force. If $x(t)$ denotes the physical quantity of interest and if $f(t)$ is the driving force, then in a linear system, an admittance function $\psi(\omega)$ can be defined by the relationship

$$x(\omega) = \psi(\omega) f(\omega), \quad (29)$$

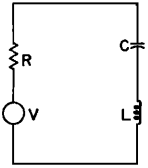
where $x(\omega)$ and $f(\omega)$ are the Fourier transforms of $x(t)$

⁴⁰ H. B. Callen and T. A. Welton, Phys. Rev. **83**, 34–40 (1951); W. Bernard and H. B. Callen, Rev. Mod. Phys. **31**, 1017–1044 (1959).

⁴¹ R. Kubo, J. Phys. Soc. Japan **12**, 570–586 (1957).



(a)



(b)

FIG. 1. Electrical circuit equivalent of an acoustic mode in a nonrelaxing fluid.

and $f(t)$, respectively. For systems involving dissipation $\psi(\omega)$ is complex, i.e.,

$$\psi(\omega) = \psi'(\omega) + i\psi''(\omega). \quad (30)$$

For example, in the system described by Eq. 28, the admittance function is

$$\psi(\omega) = [(-\rho_0 \omega^2 / \kappa^2 + M_0) - i\omega \eta_0]^{-1}.$$

We can formally regard the spontaneous fluctuations of the quantity x as the result of random forces f . The power spectra of x and f are related, from Eq. 29, by

$$\langle x^2(\omega) \rangle = |\psi(\omega)|^2 \langle f^2(\omega) \rangle \quad (31)$$

and are individually given by

$$\langle x^2(\omega) \rangle = \text{const} \psi''(\omega) / \omega \quad (32)$$

and

$$\langle f^2(\omega) \rangle = \text{const} \psi''(\omega) / \omega |\psi(\omega)|^2. \quad (33)$$

The constant can be determined by applying equipartition to the complete mean square of the fluctuating quantity

$$\langle x^2 \rangle = 2 \int_0^\infty d\omega \langle x^2(\omega) \rangle. \quad (34)$$

Perhaps the most familiar application of this analysis is the problem of thermally induced noise in an electrical circuit. Piercy⁴² has pointed out the utility of considering the problem of density fluctuations in terms of the analogous electrical problem. The technique of light scattering selects (through the Bragg condition) a particular acoustic mode of the system of wave vector κ . The time evolution of this mode is specified by Eq. 28. Piercy replaces this with an electrical model, given in Fig. 1(a), that obeys a corresponding differential equation:

$$L \left(\frac{\partial^2 q}{\partial t^2} \right) + R \left(\frac{\partial q}{\partial t} \right) + \frac{q}{C} = 0, \quad (35)$$

⁴² J. E. Piercy (to be published).

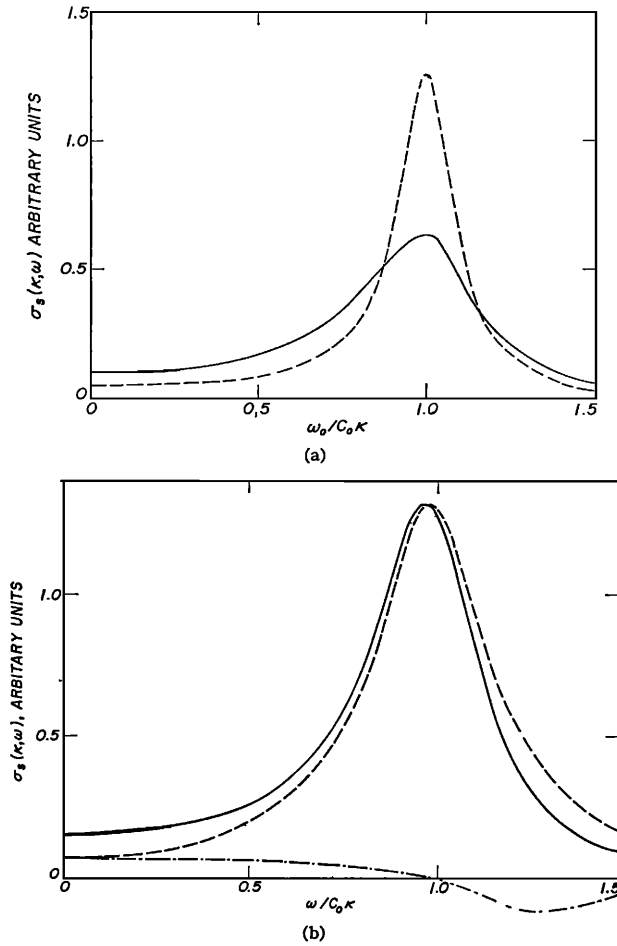


FIG. 2. The adiabatic part of the scattered spectrum for a non-relaxing fluid. (a) The effect of increasing η_0 is shown; the solid curve corresponds to a fluid having twice the viscosity of the one giving the dashed curve. (b) The synthesis of the spectrum from the terms in Eq. 39 is shown. The Lorentzian terms give the dashed curve (---) while the contribution of the last two terms is shown in the "dash-dot" curve.

where the inductance L , resistance R , capacitance C , and charge q are formally equivalent to (ρ_0/κ^2) , η_0 , $1/M_0$, and $\rho_s(\kappa, t)$, respectively. Spontaneous fluctuations in the charge q can be regarded as being produced by a noise voltage generated by the thermal motion of electrons in the resistor. Nyquist's theorem⁴³ allows one to describe the effect of this noise by a voltage source V in series with the resistor as shown in Fig. 1(b). The spectrum of the voltage source is given by the well-known expression

$$\langle V^2(\omega) \rangle = 4k_B T R. \quad (36)$$

Note that this follows from Eq. 33 using the admittance function

$$\psi(\omega) = [(-\omega^2 L + 1/C) - i\omega R]^{-1} = \frac{(-\omega^2 L + 1/C) + i\omega R}{(\omega R)^2 + (\omega^2 L - 1/C)^2} \quad (37)$$

⁴³ H. Nyquist, Phys. Rev. 32, 110-113 (1928).

obtained from Eq. 35. The spectrum of charge fluctuations can be found from either Eq. 31 or 32:

$$\langle q^2(\omega) \rangle = \frac{4k_B T R}{(\omega R)^2 + (\omega^2 L - 1/C)^2}. \quad (38)$$

This is just the form given in Eq. 25 for $\langle \rho_s^2(\kappa, \omega) \rangle$.

Observe that it is the dissipative element, the viscosity, here analogous to the resistance R , that connects the particular acoustic mode under consideration with all other modes of the system. It represents that process through which the ordered energy of the particular acoustic mode is coupled to the disordered energy of the thermal bath, i.e., to the totality of all other modes of the system. Increasing the viscosity (or resistance) both strengthens this coupling and increases the damping. This is illustrated in Fig. 2(a).

Returning to Eq. 25, some insight into the nature of the Brillouin doublet can be gotten by expanding $\sigma_s(\kappa, \omega)$ in partial fractions:

$$\sigma_s(\kappa, \omega) = \frac{1}{\gamma} \left\{ \frac{\Gamma}{\Gamma^2 + (\omega - \omega_4)^2} + \frac{\Gamma}{\Gamma^2 + (\omega + \omega_4)^2} - \left(\frac{\Gamma}{\omega_4} \right) \frac{\omega - \omega_4}{\Gamma^2 + (\omega - \omega_4)^2} + \left(\frac{\Gamma}{\omega_4} \right) \frac{\omega + \omega_4}{\Gamma^2 + (\omega + \omega_4)^2} \right\}, \quad (39)$$

where ω_4 is the free oscillation frequency of the damped system:

$$\omega_4 = \omega_0 \left[1 - \frac{1}{4} (\omega_0 \eta_0 / M_0)^2 \right]^{1/2}. \quad (40)$$

The first two terms are Lorentzian lines centered at $\pm \omega_4$ of half-width Γ . These two terms give the result obtained by Mountain²⁷ for nonrelaxing fluids if ω_4 is replaced by ω_0 . Two additional terms are seen to be present; the origin of these lies in the nature of the initial conditions imposed on the system. While not explicitly identified as such, these terms are present in the exact Mountain result for the case of single relaxation²⁸ when the limit $\tau \rightarrow 0$ is taken there. The first two terms are the Fourier transform of a correlation function

$$F_1(\kappa, t) = F(\kappa) e^{-\Gamma|t|} \cos \omega_4 t. \quad (41)$$

The absolute value of t must be taken in the exponential to take care of the required symmetry. Equation 11 requires, however, that the density correlation function have a zero derivative at $t=0$. To achieve this, it is necessary to add to $F_1(\kappa, t)$ a term in $\sin \omega_4 t$; specifically, it is necessary to add the term

$$F_2(\kappa, t) = F(\kappa) (\Gamma/\omega_4) e^{-\Gamma|t|} \sin \omega_4 t. \quad (42)$$

The Fourier transform of this gives the additional terms present in Eq. 39 and the function

$$F(\kappa, t) = F_1(\kappa, t) + F_2(\kappa, t) \quad (43)$$

satisfies the initial condition $(\partial F / \partial t)|_{t=0} = 0$ imposed on the system. Thus $F(\kappa, t)$, not $F_1(\kappa, t)$ is the correlation

function for the κ th Fourier component of the density fluctuation.

The last two terms in Eq. 39 are relatively small; the first two, on the other hand, show pronounced peaks at $\omega = \pm\omega_4$ where ω_4 is given by Eq. 40. The effects of the last two terms are illustrated in Fig. 2(b). One is to introduce asymmetry about $\omega = \pm\omega_4$ into the Brillouin lines; the second is to shift the peaks in the spectrum from the Lorentzian peaks at $\omega_4 = \pm\omega_0[1 - \frac{1}{4}(\omega_0\eta_0/M_0)^2]^{\frac{1}{2}}$ to positions nearer the center ($\omega = 0$) of the spectrum, given by Eq. 27.

One more point may be made concerning the spectral distribution given by Eq. 39. When it is integrated over frequency to obtain the Brillouin intensity, the first two terms give π/γ , while the integral of the last two terms vanishes. The ratio of Rayleigh to Brillouin intensity is therefore $\gamma-1$, which is the Landau-Placzek result. The two "new" terms in Eq. 39 do not contribute any additional intensity. Of course, extreme care must be taken in attempting to experimentally evaluate this ratio, especially if the Rayleigh intensity is relatively small. In water at 4°C, for instance, $\gamma=1$ so that the Rayleigh line should be absent. The above theory predicts that the intensity in the neighborhood of $\omega=0$ is twice that found by assuming Lorentzian functions to describe the Brillouin lines. In this case, however, the scattering due to temperature fluctuations, which we have neglected, may mask the effect. Experiments to date have been indecisive on this point.⁴⁴

If relaxational effects are present in the liquid it is necessary to modify the hydrodynamic equations (12-14) to account for them. If the process is one in which only a relaxing isothermal compressibility is present, the modification can be effected simply by adding to the right side of Eq. 13 the term

$$\frac{1}{\rho_0} \int_0^t dt' \eta(t-t') \nabla[\nabla \cdot \mathbf{u}(t')], \quad (44)$$

which accounts for a frequency-dependent shear and/or structural bulk viscosity. The after effect function $\eta(t)$ describes the viscoelastic force at time t after the application of stress. In a two-state model, for example, the term (Eq. 44) is sufficient to describe the relaxation if the two states differ only in volume, but not in enthalpy. If an enthalpy difference must also be considered, i.e., if thermal relaxation effects are to be considered, one must also allow for the frequency dependence of the specific heat C_v and the expansion coefficient φ . If, however, the time scale of energy transport is long compared with the thermal relaxation time this dependence can be ignored in Eq. 14. Additionally, Zwanzig⁴⁵ has shown that in this limit φ and C_v can also be assigned their static values in Eq. 13, the effect of the relaxation being formally contained in the integral term,

Eq. 44. Thus the relaxing viscosity described by this term can be interpreted as any adiabatic loss mechanism in the fluid. In what follows, we confine ourselves to the weak coupling, i.e., small λ limit; if the relaxation times become too long, our treatment will be inadequate; a reasonable extension of our result might be obtained by employing a two-temperature model similar to that used by Sugarawa and Yip.³⁵ The inclusion of the term (Eq. 44) in the Navier-Stokes equation means that the behavior of the system for $t>0$ is specified by the form of $\eta(t)$ in addition to the usual parameters governing a nonrelaxing medium. This correlation function is defined such that its Fourier cosine transform gives the frequency-dependent viscosity:

$$\eta(\omega) = \int_0^\infty dt \eta(t) \cos \omega t. \quad (45)$$

The quantity η_0 in Eq. 13 is now to be taken as the nonrelaxing part of the total viscoelasticity. The spectrum can then be calculated either in the manner described in the discussion preceding Eq. 16 or by employing the fluctuation-dissipation theorem. In the limit of small thermal conductivity,

$$(\lambda \kappa^2 / \rho_0 C_p) \tau \ll 1, \quad (46)$$

where τ is some characteristic or average relaxation time, the spectrum is

$$\sigma(\kappa, \omega) = \left(1 - \frac{1}{\gamma}\right) \frac{2(\lambda \kappa^2 / \rho_0 C_p)}{(\lambda \kappa^2 / \rho_0 C_p)^2 + \omega^2} + \left(\frac{2M_0}{\gamma}\right) \frac{\eta(\omega) + \eta_0}{\{\omega[\eta(\omega) + \eta_0]\}^2 + \{\rho_0 \omega^2 / \kappa^2 - M_0 - \omega \zeta(\omega)\}^2}. \quad (47)$$

In this, $\zeta(\omega)$ is the sine transform of $\eta(t)$:

$$\zeta(\omega) = \int_0^\infty dt \eta(t) \sin \omega t. \quad (48)$$

For a relaxing system, $\eta(t)$ is expressible as

$$\eta(t) = M_r \int_0^\infty d\tau g(\tau) e^{-t/\tau} \quad (49)$$

where $g(\tau)$ is the distribution function describing the relaxation time spectrum. If we set

$$\int_0^\infty d\tau g(\tau) = 1, \quad (50)$$

then M_r is the relaxational part of the longitudinal modulus.⁸ For example, in liquids where the dispersion is due to a relaxing internal specific heat C_i , M_r takes the form

$$M_r = \rho_0 c_0^2 \frac{(C_p - C_v C_i)}{(C_v - C_i) C_p}; \quad (51)$$

⁴⁴ C. L. O'Connor and J. Schlupf (to be published).

⁴⁵ R. Zwanzig, J. Chem. Phys. 43, 714-720 (1965).

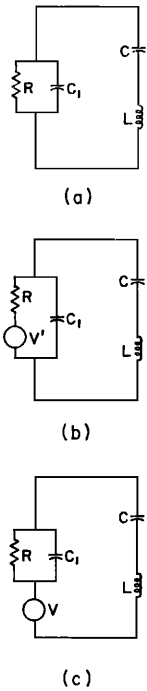


FIG. 3. Electrical equivalent of an acoustical mode for a liquid possessing a single relaxation time.

if it is the shear and compressional moduli that are relaxing

$$M_r = K_r + \frac{4}{3}G_\infty, \quad (52)$$

where K_r is the relaxational modulus of compression and G_∞ is the high-frequency limit of the shear rigidity. In any case, we may make the identification

$$M_r = \rho_0(c_\infty^2 - c_0^2), \quad (53)$$

where c_∞ is the infinite frequency sound velocity. We now recognize the quantity $\omega\zeta(\omega)$ appearing in Eq. 47 as the relaxing part of the longitudinal modulus. That is, from Eqs. 48 and 49

$$\omega\zeta(\omega) = M_r \int_0^\infty d\tau g(\tau) \frac{\omega^2 \tau^2}{1 + \omega^2 \tau^2}, \quad (54)$$

so that the spectral distribution function is given by

$$\sigma(\kappa, \omega) = \left(1 - \frac{1}{\gamma}\right) \frac{2(\lambda\kappa^2/\rho_0 C_p)}{(\lambda\kappa^2/\rho_0 C_p)^2 + \omega^2} + \left(\frac{2M_0}{\gamma}\right) \frac{\eta(\omega) + \eta_0}{\{\omega[\eta(\omega) + \eta_0]\}^2 + \{\rho_0\omega^2/\kappa^2 - M'(\omega)\}^2}, \quad (55)$$

where $M'(\omega)$ is the longitudinal storage modulus $M_0 + \omega\zeta(\omega)$. Again we separate the second term, denoted by $\sigma_s(\kappa, \omega)$, for special consideration. (The first term again represents the usual Rayleigh line). For single relaxation behavior

$$\eta(\omega) = \rho_0(c_\infty^2 - c_0^2)\tau/(1 + \omega^2\tau^2), \quad (56)$$

$$M'(\omega) = \rho_0(c_0^2 + c_\infty^2\omega^2\tau^2)/(1 + \omega^2\tau^2); \quad (57)$$

the “adiabatic” portion of the spectrum is then

$$\sigma_s(\kappa, \omega) = \left(\frac{2\rho_0 c_0^2}{\gamma}\right) \times \frac{\eta_0 + \rho_0(c_\infty^2 - c_0^2)\tau/(1 + \omega^2\tau^2)}{\left\{\omega\left(\eta_0 + \rho_0\frac{(c_\infty^2 - c_0^2)\tau}{1 + \omega^2\tau^2}\right)\right\}^2 + \left\{\rho_0\frac{\omega^2}{\kappa^2} - \frac{c_0^2 + c_\infty^2\omega^2\tau^2}{1 + \omega^2\tau^2}\right\}^2}. \quad (58)$$

This agrees with the spectral distribution formula obtained by Mountain²⁸ when the limit $\lambda \rightarrow 0$ is taken in his result.

The electrical circuit analogy, previously employed to describe a nonrelaxing system, has also been used by Piercy in the present case.⁴² For simplicity, we assume $\eta_0 = 0$; the corresponding electrical network is then shown in Fig. 3(a). The equivalence of this circuit to the single relaxation fluid can be shown by comparing the differential equation for charge in the circuit to that obtained for $\rho_s(\kappa, t)$ by setting

$$\eta(t) = M_r e^{-t/\tau}$$

in Eq. 44 and combining the resulting hydrodynamic equations. The analogous quantities are just L and ρ_0/κ^2 , R and $M_r\tau$, C_1 and $1/M_r$, and C and $1/M_0$. The spectrum of the driving noise voltage in this case is again $4k_B TR$ when the generator is placed as shown in Fig. 3(b). For the purpose of employing Eq. 31 to determine $\langle q^2(\omega) \rangle$, it is more convenient to place the generator as shown in Fig. 3(c). In this case the driving voltage spectrum can be shown by simple circuit considerations to be given by

$$\langle V^2(\omega) \rangle = 4k_B TR'(\omega) = \frac{4k_B TR}{1 + \omega^2 R^2 C_1^2}, \quad (59)$$

where $R'(\omega)$ is the effective series resistance equivalent of the parallel RC_1 combination. The driving voltage spectrum is seen to be the “white noise” thermally induced voltage in R , i.e., $4k_B TR$, filtered by the low-pass RC_1 combination. The charge spectrum follows immediately and is given by

$$\langle q^2(\omega) \rangle = 4k_B T \times \frac{R/(1 + \omega^2 R^2 C_1^2)}{(\omega R)^2 + \left(-\omega^2 L + \frac{1}{C} + \frac{1}{C_1} \frac{\omega^2 R^2 C_1^2}{1 + \omega^2 R^2 C_1^2}\right)^2}. \quad (60)$$

When viewed in this way, the unshifted component of the adiabatic light scattering spectrum found by Mountain is seen to be chiefly the result of the decreasing driving voltage in Eq. 59 as the frequency increases. Of course, other factors—namely, the variation of $\eta(\omega)$ and $M'(\omega)$ in the denominator of Eq. 60—also contribute to this “Mountain line,” but its presence and shape are principally determined by the numerator which represents the force spectrum.

The case of more than one relaxation time requires generalization of the simple circuit of Fig. 3(a). If there are N discrete relaxation times $\tau_1, \tau_2, \dots, \tau_N$, then N parallel resistance-capacitance combinations must be used in place of the combination RC_1 . The identifications are similar to those used above with $R_1C_1 = \tau_1$, $R_2C_2 = \tau_2$, $\dots$, $R_NC_N = \tau_N$. For a continuous relaxation time spectrum, a distribution of RC combinations can be envisioned to describe the process.

Returning now to our discussion of Eq. 55 we shall retain the assumption $\eta_0 = 0$. The case of an apparently frequency-independent viscosity, in addition to the relaxational viscosity, can be handled formally by including in $\eta(\omega)$ a term with some very short relaxation time. For such a case

$$\sigma_s(\kappa, \omega) = \left(\frac{2\rho_0 c_0^2}{\gamma} \right) \frac{\eta(\omega)}{[\omega\eta(\omega)]^2 + [\rho_0\omega^2/\kappa^2 - M'(\omega)]^2}. \quad (61)$$

The shape of this spectrum for various values of $c_0\kappa\bar{\tau}$ and a dispersion $(c_\infty^2 - c_0^2)/c_0^2 = 1.18$ is shown in Figs. 4 and 5. Both a single relaxation time and an asymmetric distribution of relaxation times (described by a Cole-Davidson⁴⁶ function of width parameter 0.44) are illustrated. The quantity $\bar{\tau}$ is the average relaxation time of the distribution defined by

$$\bar{\tau} = \int_0^\infty d\tau \tau g(\tau). \quad (62)$$

In order to examine $\sigma_s(\kappa, \omega)$ as was done for the non-relaxing case, the distribution of relaxation times $g(\tau)$

$$\sigma_s(\kappa, \omega) = \frac{(c_\infty^2 - c_0^2)\kappa^2 - [(\omega_B/c_0\kappa)^2 - 1][c_0^4\kappa^4/\omega_B^4\tau^2 + c_0^2\kappa^2(1 - 1/\gamma)]}{(c_0^2\kappa^2/\omega_B^2\tau)^2 + \omega_B^2} \frac{2\Gamma_M}{\Gamma_M^2 + \omega^2} + \frac{[1 - (c_0\kappa/\omega_B)^2(1 - 1/\gamma)][\omega_B^2 + c_0^2\kappa^2/\omega_B^2\tau^2] - (c_\infty^2 - c_0^2)\kappa^2}{(c_0^2\kappa^2/\omega_B^2\tau)^2 + \omega_B^2} \left\{ \frac{\Gamma_B}{\Gamma_B^2 + (\omega - \omega_B)^2} + \frac{\Gamma_B}{\Gamma_B^2 + (\omega + \omega_B)^2} \right\}. \quad (64)$$

The quantities ω_B , Γ_B , and Γ_M satisfy the equations $\omega_B^2 = M'(\omega_B)\kappa^2/\rho_0$, i.e.,

$$\omega_B^2 = \frac{1}{2\tau^2} \{ (c_\infty^2\kappa^2\tau^2 - 1) + [(c_\infty^2\kappa^2\tau^2 - 1)^2 + 4c_0^2\kappa^2\tau^2]^{1/2} \}, \quad (65)$$

$$\Gamma_B = (c_\infty^2 - c_0^2)\kappa^2\tau / (1 + \omega_B^2\tau^2)^\omega \quad (66)$$

and

$$\Gamma_M = c_0^2\kappa^2/\omega_B^2\tau. \quad (67)$$

The restriction placed on the magnitude of the dispersion is equivalent to requiring that the damping be small, i.e., $\Gamma_B \ll \omega_B$. In such instances Eqs. 64-67 give a reasonably accurate representation of the spectrum near the Brillouin peak. However, even for relatively small values of dispersion, it is not possible to use this

⁴⁶ D. W. Davidson and R. H. Cole, J. Chem. Phys. **19**, 1484-1490 (1951).

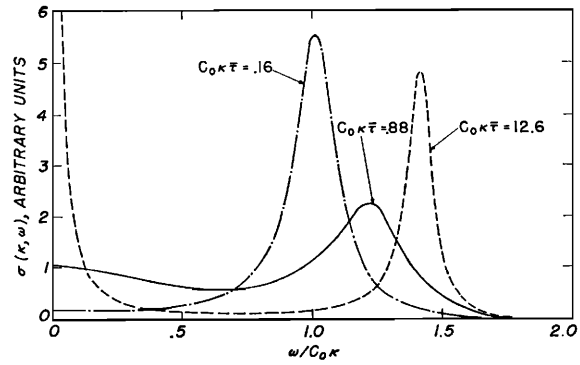


FIG. 4. Adiabatic spectra of light scattered from a liquid possessing a Cole-Davidson distribution of relaxation times (width parameter=0.44). These curves were computed for a dispersion $(c_\infty^2 - c_0^2)/c_0^2 = 1.18$.

must be known, perhaps from ultrasonic experiments. If it is not known, the form of σ_s given by Eq. 61 contains too many parameters to be useful in the analysis of experimental data. Certain approximations can be introduced to make the expression more amenable to interpretation.

In one fairly special case, Mountain²⁸ has obtained such an approximate result. For fluids exhibiting a single relaxation time τ and small dispersion, i.e.,

$$(c_\infty^2 - c_0^2)/c_0^2 \ll 1, \quad (63)$$

the viscoelastic portion of the spectrum is the sum of three Lorentzian lines

approximate result to obtain accurate relaxational data. The difficulty arises when one attempts to consider such quantities as $M' - M_0$ for values of $\omega_B\tau$ on the order of unity. Small errors present in the determination of M' will be greatly magnified when the static value M_0 is subtracted from it. Therefore, to obtain accurate data in the dispersion region it is necessary either to improve the above approximate result, or to return to the exact expression, Eq. 61, and attempt a curve-fitting procedure for the spectrum using τ as an adjustable parameter.

In attempting to obtain a best fit of the calculated spectrum to the experimentally determined one, it is necessary to convolve the calculated spectral function with an instrumental response profile to obtain a function that can be compared with experiment. This function will be sensitive to any "noise" on the instru-

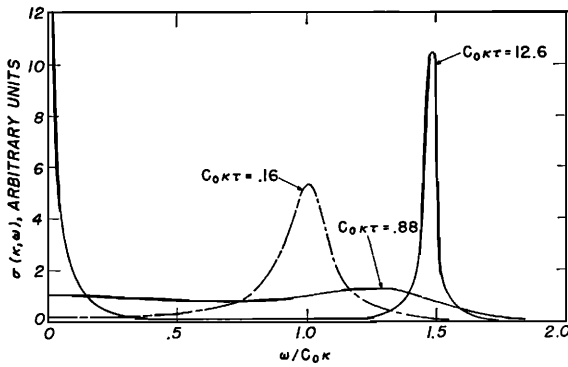


FIG. 5. Adiabatic spectra of light scattered from a fluid possessing a single relaxation time and a dispersion $(c_\infty^2 - c_0^2)/c_0^2 = 1.18$.

mental profile, especially in regions away from the principal maxima. Despite these difficulties, Stoicheff⁷ has used this approach to interpret spectra for carbon tetrachloride. In so doing, he obtains a value of τ in disagreement with data reported by other investigators.^{4,47} In view of this discrepancy and the insensitivity of this method of obtaining the exact value of τ , and due to the possibility of a distribution of relaxation times, the usefulness of this curve-fitting technique is still open to question.

Returning to Eq. 64, it is possible to improve the approximation slightly insofar as the values of the Brillouin shift frequency ω_B , the half-width of the Brillouin line Γ_B , and the half-width of the Mountain line Γ_M are concerned. The improved values can then be used in place of those given by Eqs. 65 and 66 to obtain a more accurate spectral distribution function. This approach enables one to obtain significant information in experimental situations where only the outstanding features such as the position of the shifted maxima of the spectrum are accessible. In the small thermal conductivity limit ω_B , Γ_B , and Γ_M can be found from

$$s^3 + s^2/\tau + c_\infty^2 \kappa^2 s + c_0^2 \kappa^2/\tau = (s + \Gamma_M)(s^2 + 2\Gamma_B s + \omega_B^2 + \Gamma_B^2). \quad (68)$$

The solutions are

$$\Gamma_M = (1/3\tau)\{1 - [(B-A)^{1/3} + (B+A)^{1/3}]\}, \quad (69)$$

$$\Gamma_B = (1/3\tau)\{1 + \frac{1}{2}[(B-A)^{1/3} + (B+A)^{1/3}]\}; \quad (70)$$

and

$$\omega_B = (\sqrt{3}/6\tau)\{(B-A)^{1/3} - (B+A)^{1/3}\}. \quad (71)$$

Here

$$A = 1 - 36(c_\infty^2 - c_0^2)\kappa^2\tau^2 + \frac{1}{4}(81c_0^4 - 27c_\infty^4 - 486c_0^2c_\infty^2)\kappa^4\tau^4 + 27c_\infty^6\kappa^6\tau^6 \quad (72)$$

and

$$B = -1 + \frac{1}{2}(9c_\infty^2 - 27c_0^2)\kappa^2\tau^2. \quad (73)$$

⁴⁷ E. F. Carome, S. P. Singal, and C. R. Kunsitis (to be published).

These equations relate the observable quantities to "fundamental" properties of the medium. Their use in interpreting the results of an experiment is quite limited, for in any case in which Eq. 64 may be used with confidence Eqs. 65-67 may also be used. The differences are just too small to be seen as long as $\Gamma_B \ll \omega_B$. One interesting result can nevertheless be derived from Eqs. 69 and 70. Comparing these shows that

$$\Gamma_M + 2\Gamma_B = 1/\tau. \quad (74)$$

This simple formula suggests that a density fluctuation in a fluid can be envisioned as being composed of an oscillatory part and a nonoscillatory purely decaying part. The sum of the rates at which each of these is damped gives the macroscopic relaxation time. A similar division of the density fluctuations has been proposed by Pinnow *et al.*⁴⁸ to account for the change in intensity of the Brillouin lines as the temperature, and hence the relaxation time, of a liquid is varied.

The actual use of Eq. 74 to determine τ is quite difficult in most experimental situations. The problem arises from the fact that if $\omega_B\tau \ll 1$, specifically, the case in which Eq. 74 would be of most use, the intensity in the Mountain line is quite small. And although $\Gamma_M \ll \Gamma_B$, one finds that the differential intensity of the Mountain line for $|\omega| - \omega_B > 0$ is always less than that of the Brillouin line. If $\omega_B\tau \approx 1$, Eq. 74 describing the spectrum as the sum of three Lorentzians may not be adequate. Even if it can be used, there is the nontrivial difficulty of separating the two lines in the observed spectrum. Moreover, Γ_B may not be negligible in Eq. 74 so that the difficult experimental problem of measuring the Brillouin half-width appears. While none of these present insurmountable barriers, they impose severe limitations on the precision to which τ can be experimentally found by this method.

The one case in which the Mountain line provides the relaxation time more or less directly occurs when $\omega_B\tau$ is significantly greater than unity, but is not so large that the small λ approximation is invalidated. Practically, this limits the technique to fluids in which the relaxation time lies in the range 10^{-8} - 10^{-10} sec. The use of light scattering experiments to study such relatively long times seems unusual; the usefulness of the technique here stems from the fact that τ is obtained directly from a single measurement. This is in contrast to the ultrasonic method of determining the relaxation time which requires many measurements at different frequencies.

The actual calculation of the relaxation time in this case from the light spectrum is done by realizing that, from Eq. 67, the Mountain half-width is $\Gamma_M = c_0^2/c_\infty^2\tau$, and the position of the Brillouin peak yields

$$c_\infty = \omega_B/\kappa, \quad (75)$$

so that τ may be found if c_0 is known. In this case, one may also obtain τ from Eq. 75 and the fundamental

⁴⁸ D. A. Pinnow, S. J. Candau, J. T. LaMacchia, and T. A. Litovitz, J. Acoust. Soc. Am. 43, 131-142 (1967).

viscoelastic relation

$$\lim_{\omega \rightarrow 0} \eta(\omega) = \frac{4}{3}\eta_s + \eta_v = \rho_0(c_\infty^2 - c_0^2)\tau. \quad (76)$$

These provide τ if the other parameters are known.

The static shear viscosity η_s can be measured using standard viscometric techniques and the static volume viscosity η_v and c_0 can be found from low-frequency ultrasonic absorption and velocity data. Experimental agreement on the values of τ obtained using these two approaches would provide some verification of the assumption of single relaxation underlying the whole treatment.

In the case of multiple relaxation, Nichols and Carome⁴⁹ have recently adapted Mountain's calculation to include the effect of two discrete relaxation times and, in principle, their result can be extended to apply to any finite number of discrete relaxation times. If a continuous distribution of times is used to describe the relaxation spectrum, a simple factoring of the dispersion relation, as was done for a single time in Eq. 68, is impossible. The best procedure seems to be to develop some approximation scheme to give the desired viscoelastic quantities in terms of the outstanding features of the scattered light spectrum.

II. APPROXIMATE TREATMENT OF THE SPECTRUM FOR A RELAXING LIQUID

In many experimental situations, especially those in which the liquid exhibits a distribution of relaxation times, attempting to represent the scattered light spectrum by adjusting various parameters in Eq. 61 provides little useful information. This is because the experimental data can often be made to fit this equation quite acceptably within rather wide variations of the adjustable parameters. The detailed shape of the spectrum is relatively insensitive to the exact nature of the distribution of relaxation times. If Eq. 61 is convolved with an instrumental response function and the result compared with an experimentally obtained spectrum, experimental inaccuracies will prohibit choosing one distribution of relaxation times in preference to another. In fact, one finds that the only obvious criteria for distinguishing distributions are the position of the Brillouin peak and the half-width of the Brillouin line.

If only these features of $\sigma_s(\kappa, \omega)$ are sensitive to the relaxational spectrum of the liquid under investigation, then it is just these features that should be measured to obtain relaxational data. (The intensity of the Brillouin line can also provide useful relaxational information, and this interesting aspect of Brillouin spectra has been discussed by Pinnow.⁴⁸) Measuring only a few features of the spectrum, however, is equivalent to ignoring most of $\sigma_s(\kappa, \omega)$, and therefore one must be content with obtaining something less than all the relaxational characteristics from a single measurement. The situation

⁴⁹ W. H. Nichols and E. F. Carome (to be published).

may be clarified somewhat by examining the approximate spectral function of Eq. 64. Here measurement of the Brillouin shift ω_B and half-width Γ_B provide (approximately) the sound velocity c and the sound absorption coefficient α at the frequency ω_B . The necessary formulas are

$$c \approx \omega_B/\kappa \quad (77)$$

and

$$\alpha \approx \Gamma_B \kappa / \omega_B. \quad (78)$$

If these are measured at various values of ω_B , then the desired information can be extracted from the curves of c and α versus ω_B . The variation of ω_B is accomplished by changing κ , that is by making observations at several scattering angles. Practically, this limits the range of ω_B to about one frequency decade; $\omega_B/2\pi$ may be varied from approximately 0.7 to 7.0 GHz. In certain cases, however, one can construct reduced functions from combinations of the measured variables; a quantity is a reduced function if its frequency and temperature behavior enters only through the product $\omega\tau$. In these instances, the "frequency" range of the experiment can be extended considerably by changing the temperature and therefore changing τ ; the plots of the reduced functions versus $\omega\tau$ then replace those of c and α versus ω .

In the case at hand, we attempt to construct such reduced functions of the product $\omega\tau$ and show how these functions are related to the observable quantities. The functions chosen are those commonly used in acoustic propagation experiments. The value of this approach is obvious; light-scattering data may be compared directly with acoustical data resulting in a further extension of the available frequency range. The functions commonly used in acoustical studies are directly related to the real and imaginary parts of the longitudinal modulus at the angular frequency ω_A , M' , and M'' , respectively:

$$\Re' = \frac{M' - M_0}{M_\tau} = \int_0^\infty d\tau g(\tau) \frac{\omega_A^2 \tau^2}{1 + \omega_A^2 \tau^2} \quad (79)$$

and

$$\Re'' = \frac{M''}{M_\tau} = \frac{\omega_A \eta(\omega_A)}{M_\tau} = \int_0^\infty d\tau g(\tau) \frac{\omega_A \tau}{1 + \omega_A^2 \tau^2}. \quad (80)$$

In these, $\Re'$ and $\Re''$ are the reduced storage and loss moduli, respectively. In an ultrasonic experiment, M' and M'' are related to the propagation characteristics by the well-known equations

$$M' = \rho_0 \frac{\omega_A^2}{k^2} \left\{ \frac{1 - (\alpha/k)^2}{[1 + (\alpha/k)^2]^2} \right\}, \quad (81)$$

and

$$M'' = \rho_0 \frac{\omega_A^2}{k^2} \left\{ \frac{2(\alpha/k)}{[1 + (\alpha/k)^2]^2} \right\}, \quad (82)$$

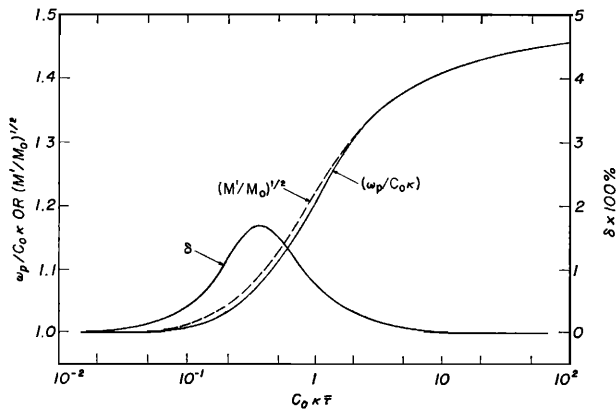


FIG. 6. The dependence of $\omega_p/c_0\kappa$ and $[M'(\omega_p)/M_0]^{1/2}$ on $c_0\kappa\tau$ assuming a Cole-Davidson distribution of relaxation times (width parameter=0.44) and a dispersion $(c_\infty^2 - c_0^2)/c_0^2 = 1.18$. The peaked curve shows δ vs $c_0\kappa\tau$ for these parameters.

where k is the wave vector of the sonic wave. The problem is now to relate these quantities to the parameters measured in a light scattering experiment.

A brief inspection of Eq. 61 shows that the peak in $\sigma_s(\kappa, \omega)$ should correspond approximately with the vanishing of the second term in the denominator.²⁸ Define a frequency ω' as that frequency which satisfies this condition, i.e.,

$$\omega'^2 - M'(\omega')\kappa^2/\rho_0 = 0. \quad (83)$$

However, the actual frequency at the peak is the solution of the equation

$$\partial \sigma_s(\kappa, \omega)/\partial \omega|_{\omega=\omega_p} = 0. \quad (84)$$

The exact evaluation of ω_p from this last expression is quite complicated and contributes little toward obtaining an expression for M' . We may, however, assume that ω_p is not far different from ω' and write

$$\omega_p = \omega'(1 - \delta_p). \quad (85)$$

Then expanding η and M' about ω' in powers of δ_p and dropping terms of second and higher order leads to

$$\delta_p = \left[\frac{\omega' \eta(\omega')}{2M'(\omega')} \right]^2 \frac{1 + \omega'(\partial \eta / \partial \omega') / 2\eta(\omega')}{1 - \omega'(\partial M' / \partial \omega') / M'(\omega')}, \quad (86)$$

where the notation $\partial/\partial \omega'$ implies that the differentiated function of ω is to be evaluated at ω' . The modulus at the peak, $M'(\omega_p)$, can then be expressed as

$$M'(\omega_p) = \rho_0(\omega_p/\kappa)^2(1 + 2\delta). \quad (87)$$

As before, we compute δ to first order. The result is

$$\delta = \left[\frac{\omega_p \eta(\omega_p)}{2M'(\omega_p)} \right]^2 \frac{1 + \omega_p(\partial \eta / \partial \omega_p) / 2\eta(\omega_p)}{1 - \omega_p(\partial M' / \partial \omega_p) / 2M'(\omega_p)}, \quad (88)$$

where, as above, $\partial/\partial \omega_p$ means that the differentiated function of ω is to be evaluated at ω_p . In terms of $g(\tau)$

TABLE I. Typical correction terms arising from Eq. 89 for various liquids, computed for $\omega_p\tau = 1$ assuming single relaxation. The last two columns give, respectively, the percentage corrections to the hypersonic velocity c and the longitudinal modulus M' .

Liquid	c_∞/c_0	M_r/M_0	$(c - \omega_p/\kappa)$	$(M' - \rho_0\omega_p^2/\kappa^2)$
			c	M'
Chloroform	1.12 ^a	0.25	0.2%	0.4%
Carbon tetrachloride	1.22 ^a	0.50	0.6%	1%
Benzene	1.22 ^a	0.50	0.6%	1%
Glycerol	1.48 ^b	1.18	4%	8%

^a Data taken from Ref. 8, p. 363.

^b Data taken from Ref. 48.

and M_r/M_0 ,

$$\delta = \frac{M_r}{4M_0} \frac{\int_0^\infty d\tau g(\tau) \omega_p \tau / (1 + \omega_p^2 \tau^2)}{1 + (M_r/M_0) \int_0^\infty d\tau g(\tau) \omega_p^2 \tau^2 / (1 + \omega_p^2 \tau^2)} \times \frac{\int_0^\infty d\tau g(\tau) \omega_p \tau / (1 + \omega_p^2 \tau^2)^2}{1 + \left(\frac{M_r}{M_0} \right) \int_0^\infty d\tau g(\tau) \omega_p^2 \tau^2 \left[\frac{1}{1 + \omega_p^2 \tau^2} - \frac{1}{(1 + \omega_p^2 \tau^2)^2} \right]}. \quad (89)$$

Equations 85, 87, and 89 are plotted in Figs. 6 and 7. The "correction" δ is seen to be of the order of a few percent. This represents a significant correction to be made in evaluating M' from ω_p . Typical values of δ are also shown in Table I for several types of liquids. The values here are computed for $\omega_p\tau = 1$ assuming single relaxation times. These circumstances yield values of δ near the maximum of this correction term regarded as a function of $\omega_p\tau$. In an experimental situation where $g(\tau)$ is not known, one may obtain the quantity δ by obtaining a first approximation to $g(\tau)$ by plotting $[(\rho_0\omega_p^2/\kappa^2) - M_0]/M_r$ versus $\omega_p\tau$ and fitting the resulting curve to Eq. 79. The approximate distribution function found by this procedure can then be inserted into Eq. 89 to calculate δ . Then, using Eqs. 79 and 87, a more accurate distribution $g(\tau)$ can be found. Alternatively, one may determine δ from Eq. 88 by evaluating the necessary derivatives graphically.

The real part of the longitudinal modulus can thus be obtained from the peaks of the Brillouin spectrum. The imaginary part can be related to the half-width of the resonance line. The appropriate formulas can be obtained as follows.

Consider Eq. 61 for $\sigma_s(\kappa, \omega)$. This depends upon frequency both explicitly and implicitly through the dependence of $\eta(\omega)$ and $M'(\omega)$ on frequency. Let us make this dependence explicit by defining a function

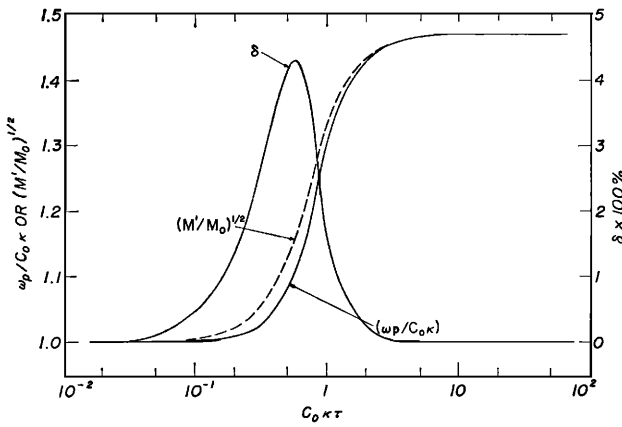


FIG. 7. The dependence of $\omega_p/c_0\kappa$ and $[M'(\omega_p)/M_0]^{1/2}$ on $c_0\kappa\tau$ assuming single relaxation and a dispersion $(c_\omega^2 - c_0^2)/c_0^2 = 1.18$. The peaked curve shows δ vs $c_0\kappa\tau$ for these parameters.

$G[\kappa, \omega, \eta(\omega), M'(\omega)]$ by

$$\sigma_s(\kappa, \omega) = G[\kappa, \omega, \eta(\omega), M'(\omega)]. \quad (90)$$

If the Brillouin line is not too broad, we may expand η and M' about ω_p to obtain a function that approximates the true spectrum in the neighborhood of ω_p . For $|\omega| - \omega_p \geq 0$, retaining only the first terms in these expansions gives quite a good approximation as is shown in Fig. 8. The half-width of the line Δ is then defined by the equation

$$\sigma(\kappa, \omega_p + \Delta) = \sigma(\kappa, \omega_p)/2. \quad (91)$$

Then, using the approximation just described in Eq. 91 and considering only values of $\Delta > 0$, this leads to

$$(\kappa^2/\rho_0)^2(\omega_p + \Delta)^2\eta(\omega_p)^2 + [(\omega_p + \Delta)^2 - M'(\omega_p)\kappa^2/\rho_0]^2 = 2\omega_p^2\eta(\omega_p)^2\kappa^4/\rho_0^2 + 2[\omega_p^2 - M'(\omega_p)\kappa^2/\rho_0]^2. \quad (92)$$

Recalling that $\omega_p\eta(\omega_p) = M''(\omega_p)$, this can be solved for the ratio M''/M' . Neglecting terms of order $(\Delta/\omega_p)^4$ and higher, this gives

$$\left[\frac{M''(\omega_p)}{M'(\omega_p)}\right]^2 \approx \left(\frac{2\Delta}{\omega_p}\right)^2 \left[1 + 3\left(\frac{\Delta}{\omega_p}\right)\right] - 4\delta\left(\frac{2\Delta}{\omega_p}\right), \quad (93)$$

where $\delta = [M'(\omega_p)\kappa^2/\rho_0 - \omega_p^2]/2\omega_p^2$ as defined in Eq. 87. Taking the root of Eq. 93 gives

$$\frac{M''(\omega_p)}{M'(\omega_p)} = \left(\frac{2\Delta}{\omega_p}\right) \left[1 + \left(\frac{3\Delta}{2\omega_p}\right) - \frac{\delta}{(\Delta/\omega_p)}\right]. \quad (94)$$

Using Eq. 88 we can write δ to first order as just $(\Delta/\omega_p)^2$. In this approximation,

$$\frac{M''(\omega_p)}{M'(\omega_p)} \approx \left(\frac{2\Delta}{\omega_p}\right) \left(1 + \frac{\Delta}{2\omega_p}\right). \quad (95)$$

This is the form most useful in the analysis of experimental data. The ratio M''/M' is a function both of the

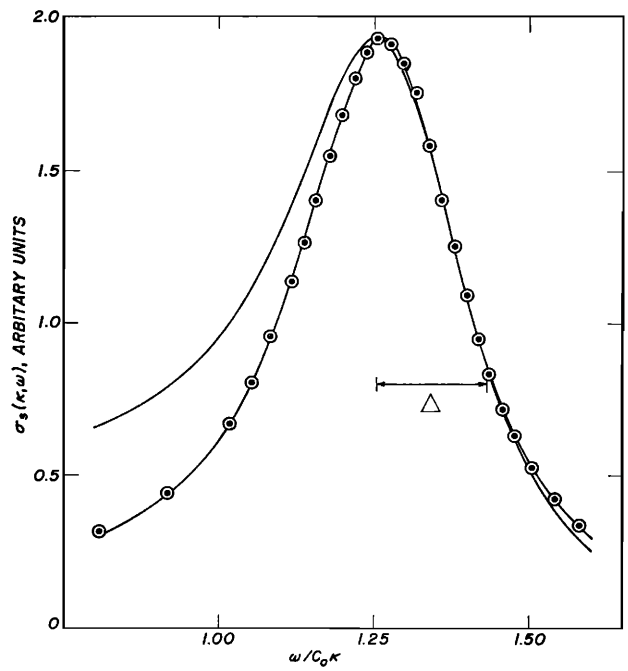


FIG. 8. The Brillouin spectrum (—) near the peak frequency ω_p for a liquid showing a dispersion $(c_\omega^2 - c_0^2)/c_0^2 = 1.18$ and a Cole- Davidson distribution (width parameter $\beta = 0.52$). The points —○— are computed from the approximate expression in Eq. 91. The quantity Δ is the half-width of the line.

product $\omega_p\tau$ and the dispersion M_r/M_0 . The approximate result obtained in Eq. 95 can be tested as follows: For a given dispersion, Δ/ω_p should be near a maximum at $\omega_p\tau \approx 1$. This is where the approximations made in deriving Eq. 95 should be worst. Therefore, assuming single relaxation (again a maximum error situation) a critical evaluation of the accuracy of Eq. 95 can be made for various values of M_r/M_0 . Such an evaluation is shown in Table II. As can be seen, the errors become rather large as the width of the Brillouin line increases. In most realistic situations, Eq. 95 provides to within 10% in the relaxation region and to much better than this outside of this region. This is comparable to the errors involved in the ultrasonic determination of M''/M' for large absorptions.

The approximations made in deriving Eq. 96 for M'' are not nearly as good as those used in obtaining Eq. 88 for M' . This is consistent with the experimental fact that it is far more difficult to obtain precise linewidth measurements than to measure the position of the peak of a line. It should be emphasized that the halfwidth Δ can only be measured after the instrumental response has been properly deconvolved. The tables of Leidecker *et al.*⁵⁰ are especially useful for this purpose when only the half-width of the line is desired.

If it is desired, one may use M' and M'' to compute the sound absorption and velocity at the frequency ω_p .

⁵⁰ H. W. Leidecker and J. T. LaMacchia, J. Acoust. Soc. Am. 43, 143-151 (1967).

TABLE II. Evaluation of M''/M' from the Brillouin scattering spectrum. The values shown are computed for the case $\omega_p\tau=1$. The percent error values listed in Column 6 are obtained from Columns 4 and 5, and represent the error made in applying the approximate relation Eq. 96 to experimental data.

$M\tau/M_0$	$g(\tau)$	Δ/ω_p	M''/M' (exact) (Eqs. 79, 80)	M''/M' (approx.) (Eq. 96)	Percent error
0.25	SRT*	0.0517	0.111	0.106	-4.5%
0.50	SRT	0.103	0.200	0.215	7.5%
1.00	SRT	0.180	0.333	0.392	18%
1.18	Cole-Davidson ($\beta=0.52$)	0.135	0.307	0.290	-5.5%

* Single relaxation time.

The required equations are

$$\alpha/k = (M'/M'')\{[1 + (M''/M')^2]^{1/2} - 1\} \quad (96)$$

and

$$c^2 = \frac{M'}{\rho_0} \frac{[1 + (\alpha/k)^2]^2}{1 - (\alpha/k)^2}. \quad (97)$$

As long as the loss is small, $\alpha/k \ll 1$, these reduce simply to

$$\frac{\alpha}{k} \approx \frac{M''}{2M'} \approx \frac{\Delta}{\omega_p} \quad (98)$$

and

$$c^2 \approx M'/\rho_0 \approx \omega_p^2/k^2. \quad (99)$$

It is therefore possible to correlate the results of light-scattering measurements with ultrasonic data.

III. SUMMARY AND CONCLUSIONS

A few additional remarks can be made in summarizing the principal results obtained. The Brillouin peaks in the scattered light spectrum have been found to be thermally driven resonance lines; the scattered light is modulated by the frequency spectrum of an acoustical mode of the system that is driven by random forces of the thermal bath. The concept of a driving force here is completely analogous to that of a "noise electromotive force" used in the analysis of Nyquist noise in an electrical circuit. That the spectrum is that of a driven system means that the shifted lines are not Lorentzian as they would be for scattering from a freely decaying system. This means that one cannot identify the

Brillouin frequency as the phase velocity times the wave vector of a temporally damped acoustic wave. However, for a simple nonrelaxing fluid it is possible to simply relate the frequency shift of the Brillouin lines to the sound velocity. The relationship is quite analogous to that connecting resonant and natural frequencies of an oscillatory system with damping.

For relaxing liquids, the interpretation of the spectrum is more complicated. Fundamentally, the reason for this is that the quantities analogous to the damping constant and natural frequency of the nonrelaxing system are here frequency dependent. The use of electrical models gives some insight into the effect of this frequency dependence. The Mountain line, for example, is produced chiefly by the relaxation of the viscosity, i.e., by the "filtering" of the driving force, at high frequencies. While an exact expression giving the spectral distribution of scattered light can be written in terms of the relaxational or viscoelastic parameters of the medium, and although the general shape of the spectrum can be understood in terms of these, the use of this function to obtain values for them from experimental spectra is quite limited. To circumvent this difficulty, approximate expressions relating the outstanding features of the spectrum to the longitudinal storage and loss moduli of the liquid have been derived.

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